

[illegible]

12. Which voltage level is commonly used for single-phase residential supply in India?
 A) 415 V B) 230 V
 C) 11 kV D) 132 kV
13. Which component allows an EV to capture and store energy that is normally lost during braking?
 A) Regenerative Braking System B) DC-DC Converter
 C) Power Electronics Controller D) Thermal System
14. In a boost converter, output voltage is _____
 A) Input Voltage / (1-D) B) D x Input Voltage
 C) (1-D) / Input Voltage D) (1-D) x Input Voltage
15. What component in a BLDC motor replaces the mechanical commutator and brushes found in a conventional DC motor?
 A) A flywheel B) An electronic controller
 C) A gearbox D) A universal joint
16. Buck converter is also called as _____
 A) Step up converter B) Step down converter
 C) Step Up - Step down converter D) Step down - step up converter
17. The property of a material which opposes the creation of magnetic flux in it is known as
 A) Reluctivity B) Magnetomotive force
 C) Permeance D) Reluctance
18. A transformer has 500 turns in the primary winding and 100 turns in the secondary winding. If the primary voltage is 220V, the secondary voltage will be
 A) 44V B) 1100V
 C) 110V D) 22V
19. Which type of pipe is used in pipe earthing
 A) Steel B) Galvanized Iron
 C) Copper D) Silver
20. The principle of operation of a DC motor is based on
 A) Kirchoff's Law B) Lenz's Law
 C) Faraday's Law of Electromagnetic Induction D) Ohm's Law

PART - B: DESCRIPTIVE ANSWER QUESTIONS

Unit – I

1. a) State and explain Ohm's law.
 b) Explain the fundamental Tie-set matrix using suitable example.
 c) Using mesh analysis calculate the current in each branch for the network shown in Fig. 1c.

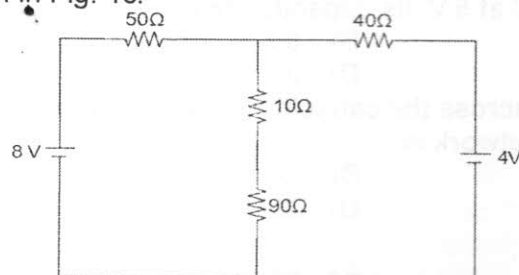


Fig. 1c

2. a) Define
 i. Node
 ii. Branch
 iii. Cut-set matrix

Marks	BT*	CO*	PO*
3	L*1	1	1,2
5	L2	1	1,2
8	L3	1	1,2
3	L1	1	1,2

- b) Draw the graph of the network shown in Fig. 2b. Also, write the incidence matrix and reduced incidence matrix.

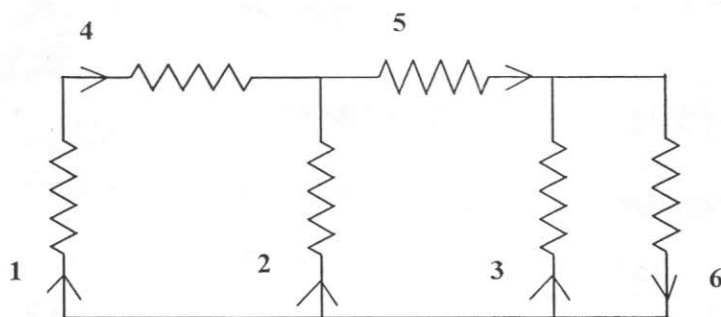


Fig. 2b

- c) Determine the node voltage V_1 and V_2 for the network shown in Fig. 2c and also calculate the current through $2\ \Omega$ resistor.

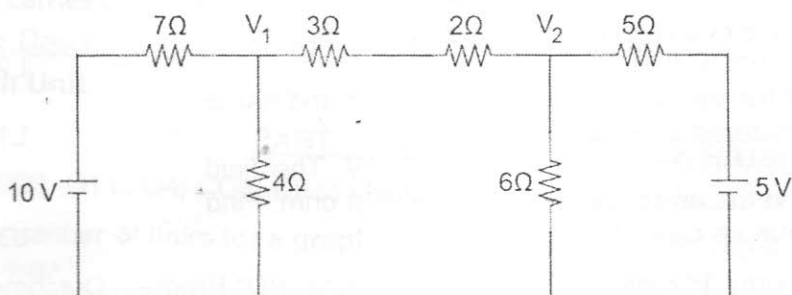


Fig. 2c

Unit – II

3. a) Define average value of an alternating quantity. Obtain an expression for average value of an alternating quantity.
- b) A circuit having resistance of $6\ \Omega$, an inductance of $0.075\ \text{Henry}$ and a capacitance of $50\ \mu\text{F}$ in series is connected across a $100\ \text{V}$, $50\ \text{Hz}$ supply. Calculate (a) the impedance (b) the current (c) the voltage across R , L and C (d) the phase difference between the current and the supply voltage.
4. a) Obtain an expression for average power consumption in a RC series circuit with neat diagram and phasor diagram.
- b) The equation of an alternating current is given by $i = 28.28 \sin(314t + 30^\circ)$. Find its RMS value, frequency and its phase angle.
- c) Explain the types of power in an electric circuit.

Unit – III

5. a) With a typical layout explain the working of hydroelectric power plant.
- b) List the advantages and disadvantages of HVAC and HVDC transmission system.
6. a) Explain with a neat schematic diagram the working of thermal power plant.
- b) Mention the voltage level and with a schematic explain the primary and secondary distribution system.

5 L2 1 1,2

8 L3 1 1,2

8 L2 2 1,2

8 L3 2 1,2

8 L2 2 1,2

5 L3 2 1,2

3 L1 2 1,2

8 L2 3 1,2

8 L1 3 1,2

8 L2 3 1,2

8 L1 3 1,2

Unit – IV

7.	a)	Explain the operation of boost converter with a neat diagram.	8	L2	4	1,2
	b)	With a neat block diagram explain the fundamental components of EVs.	8	L2	4	1,2
8.	a)	Explain the operation of BLDC motor with a suitable diagram.	8	L2	4	1,2
	b)	With a neat block diagram explain the function of UPS and list its applications.	8	L2	4	1,2

Unit – V

9.	a)	State and explain				
	i.	Self-induced e.m.f.				
	ii.	Mutually induced e.m.f.	4	L2	5	1,2
	b)	Explain the working of transformers with a suitable diagram.	4	L2	5	1,2
	c)	Outline the reasons for earthing any electrical equipment. With a neat sketch, explain the plate earthing.	8	L2	5	1,2
10.	a)	Explain the working principle of a stepper motor with a neat diagram and list its applications.	8	L2	5	1,2
	b)	List the characteristics of the materials used for fuse wires and name any two factors to be considered while selecting a fuse.	4	L1	5	1,2
	c)	The power input to a 220V D.C. shunt motor is 10 kW. The field resistance is 230 ohm and the armature resistance is 0.28 ohm. Find the input current, and armature current.	4	L3	5	1,2

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
